CINEMATCH: THE COMPLETE BEGINNER'S CODE GUIDE

A Clear, Friendly Walkthrough of Every Single File, Script, and Technology

Written for students, professors, and teammates to understand exactly how the system works without getting lost in jargon.

Project Name:	CineMatch — AI Movie Recommender & OTT Streaming Platform
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What Is in This Guide?

1. **The Big Picture:** How all the files connect like a modern digital movie studio.
2. **File-by-File Breakdown:** What each file does, plain-English analogies, and key code.
3. **The 2 Main Faces of CineMatch:** The Python Streamlit app vs. the live GitHub Pages OTT web app.
4. **How the AI Works:** TF-IDF and Cosine Similarity explained like matching flavors of ice cream.
5. **Payment & Streaming:** How the UPI QR system and video player operate safely.

1. THE BIG PICTURE: HOW EVERYTHING CONNECTS

Think of CineMatch like a **modern high-tech cinema**. To run smoothly, it needs three main departments:

• **The Storage Room (Data):** Holds the movie catalogs, descriptions, ratings, and posters (movies.csv, user_ratings.csv).

• **The Brain / Kitchen (Machine Learning & Python):** Studies movie flavors, plots, and user tastes to decide which films match (recommender.py, collaborative_filter.py).

• **The Theater Screens (User Interfaces):** Where people browse, watch trailers, buy subscriptions, and discover movies (index.html for the live web app, app.py for the Python dashboard).

File Name	Category	One-Sentence Beginner Summary
index.html	Frontend (Live)	The complete Netflix-style OTT website that visitors see and interact with.
movies.csv	Data	The movie spreadsheet containing 30 films with ratings, genres, and posters.
user_ratings.csv	Data	A table of simulated user scores used for collaborative filtering experiments.
app.py	Python App	A Python Streamlit dashboard with a movie roulette game and quiz.
recommender.py	AI Core	The Python script that calculates mathematical cosine similarity between movies.
collaborative_filter.py	AI Core	Predicts what you like by finding people who share your movie taste.
launch_public.py	Networking	Shares your local Python app to any smartphone via a Cloudflare secure link.
verify_live.py	Testing	A robot browser that tests the live website and clicks buttons automatically.
build_full_academic_report.py	Tools	Builds the formal Word and PDF reports for university submission.

2. THE DATA FILES: WHAT FEEDS THE SYSTEM

File 1: movies.csv

What is it? This is an Excel-style comma-separated table that acts as our database. It contains 30 carefully curated world-class movies.

What columns are in it?

• title: Name of the film (e.g., *Inception*, *3 Idiots*, *Dangal*).
• genre: Category (e.g., Sci-Fi, Action, Comedy, Drama).
• rating: The IMDb score out of 10 (e.g., 8.8).
• mood: Emotional vibe (e.g., *Mind-Bending* ■, *Heartwarming* ■, *Adrenaline* ■).
• stars: Leading actors (e.g., Leonardo DiCaprio, Aamir Khan).
• overview: A 2-sentence summary of the story.
• poster: Direct internet URL to official high-resolution studio poster artwork.

Simple Analogy: This is the menu of our movie restaurant. Without it, the chefs (algorithms) wouldn't know what they are serving!

File 2: user_ratings.csv

What is it? A table of 140+ simulated ratings from different users (User 1, User 2, User 3...) rating movies on a scale of 1 to 10.

Why is it needed? It trains our **Collaborative Filtering** algorithm. By seeing that 'User 1' and 'User 3' both gave 10/10 to *Interstellar* and *Inception*, the computer realizes they have identical taste and recommends other movies they each watched.

3. THE LIVE WEB PLATFORM: index.html

What is it? This is the crown jewel of the project. It is a complete, single-file Over-The-Top (OTT) streaming platform inspired by Netflix and Disney+ Hotstar. Anyone with a browser on an iPhone, Android, or laptop can open it instantly.

Why is it built as a 'Single-File App'?
Normally, websites need complicated servers, Node.js packages, and third-party script libraries. But Indian cellular carriers (like Jio and Airtel) often block third-party CDNs. To make CineMatch 100% unbreakable, the entire HTML structure, CSS styling, and JavaScript logic are bundled cleanly into one file with zero external dependencies!

What Happens Inside index.html (Section by Section):

1. **The Dark Cinema Theme (CSS):** Deep obsidian background (#07070b) with Netflix-red highlights (#e50914) and golden VIP accents (#ffd700).
2. **Hero Spotlight Billboard:** When you open the page, a massive 88vh billboard showcases a top trending movie with 1280px high-res background art and a 'Watch Now' button.
3. **Movie Carousels:** Smooth horizontally scrolling rows (Trending, Top Rated, Indian Blockbusters, Hollywood Hits).
4. **Mood Selector Quiz:** Click 'Mind-Bending' or 'Laugh Out Loud' and the catalog instantly rearranges to match your vibe.
5. **Cinema Video Modal:** Clicking any movie opens a popup cinema player. It supports multi-server switching (Server 1: YouTube HD, Server 2: Fast CDN, Server 3: Free Web Stream Aggregator).
6. **UPI Payment Gateway:** Click 'Upgrade VIP' to choose a pass (Mobile ■149, Standard ■499, VIP ■699). It creates a real dynamic QR code for Axis Bank UPI ID 7499525420@axl, accepts the demo payment, and showers the screen in confetti!
7. **Client-Side AI Engine:** Type any movie in the search box. In less than 3 milliseconds, JavaScript calculates vector overlap and tells you the top 6 films with the closest match!

4. THE MACHINE LEARNING ENGINES: HOW THE AI THINKS

File 3: recommender.py (Content-Based Filtering)

What is it? The Python engine that analyzes movie features (genre, plot words, actors, mood) to recommend similar films.
How it works (The 3-Step Recipe):
1. **Soup Creation:** For each movie, it creates a text 'soup': 'Sci-Fi Mind-Bending Leonardo DiCaprio dream thief corporate secrets'.
2. **TF-IDF Vectorizer:** Converts words into numbers. Common words like 'the' are ignored. Rare, important words like 'wormhole' or 'wrestler' get heavy mathematical scores.
3. **Cosine Similarity:** Measures the angle between two movies. If the angle is 0 degrees, the cosine is 1.0 (exact match). If the angle is 90 degrees, the cosine is 0.0 (completely different).

File 4: collaborative_filter.py (People-Like-You Filtering)

What is it? Instead of looking at movie descriptions, this script looks at *human behavior*.
Simple Analogy: If your best friend has the exact same taste in music as you, and they tell you about a new song they loved, you will probably love it too! This script mathematically calculates who your 'taste twins' are from the rating matrix.

5. THE PYTHON ANALYTICAL DASHBOARD: app.py

What is it? While index.html is designed as a public consumer streaming site, app.py is the interactive Python dashboard built using **Streamlit**.
Why did we build both?
In university evaluations, professors love to see Python data exploration! app.py lets students and professors interactively inspect similarity scores, view bar charts, spin the 'Movie Roulette' wheel, and take a 30-second personality quiz.

Key Features in app.py:

- **Movie Roulette:** Can't decide what to watch? Hit 'Spin Roulette' and it selects a certified masterpiece with balloons and confetti.
- **30-Second Quiz:** Asks 3 fun questions (Who are you with? What's your energy level? How much time do you have?) and computes your ideal movie.
- **Live Watchlist:** Add films to your personal list and export them directly to WhatsApp with 1 click.

6. NETWORKING, TESTING & AUTOMATION SCRIPTS

File 6: launch_public.py (Cloudflare Quick Tunnel)

What is it? A Python script that creates a free, secure global HTTPS internet link for your local Python Streamlit app.
Why is it amazing? Normally, to let someone on their phone view your project, you'd need to pay for cloud servers, configure routers, or port-forward. launch_public.py runs Cloudflare in the background, gives you a URL like https://something.trycloudflare.com, and generates a 1-tap WhatsApp link so you can text it to anyone!

File 7: verify_live.py (Automated Robot Tester)

What is it? An automated testing script built with **Playwright**.
How it works: It launches an invisible headless browser, visits the live website, clicks the Bank button, tests the UPI payment modal, checks mobile phone dimensions (iPhone 14 / 390px), takes 7 proof screenshots, and verifies that there are zero bugs or crashes.

File 8: build_full_academic_report.py (Report Compiler)

What is it? A Python program using python-docx and reportlab that automatically writes and styles the formal academic reports submitted to MIT-WPU professors.

7. PRESENTATION CHEAT SHEET: HOW TO EXPLAIN THIS IN 60 SECONDS

If your professor or judge asks: *'What did you build and how does it work?'*, here is the perfect beginner-friendly answer:

'We built CineMatch — an AI-powered movie recommendation and streaming platform. We combined Natural Language Processing (TF-IDF) and Cosine Similarity to mathematically understand film plots, genres, and moods. On the frontend, we built an industry-standard Netflix-style Over-The-Top web app deployed live on GitHub Pages with an embedded cinema player and an operational UPI-only subscription gateway (Axis Bank). On the backend, we built Python algorithms and a Streamlit dashboard with a movie roulette game and automated Playwright testing.'